

FORT ST JOHN, BC, Canada

WMO: 719430

Lat: 56.2467N

Lon: 120.7371W

Elev: 695

StdP: 93.25

Time Zone: -7.00 (W07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.0	-28.8	-37.2	0.1	-31.5	-34.1	0.2	-28.8	13.4	-0.7	12.0	0.6	2.0	350	0.713

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	26.2	16.1	24.5	15.3	22.9	14.5	17.5	24.1	16.3	22.8	15.3	21.4	3.8	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
14.9	11.5	19.7	13.7	10.6	18.4	12.7	10.0	17.6	51.8	24.1	48.2	23.0	45.2	21.6	22.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.4	9.4	8.3	-34.8	29.5	3.3	1.6	-37.2	30.7	-39.1	31.7	-40.9	32.6	-43.3	33.8		
			-35.0	19.0	3.2	1.4	-37.3	20.1	-39.2	20.9	-41.0	21.7	-43.3	22.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	2.3	-12.7	-9.7	-5.5	3.3	9.9	14.1	16.3	15.0	10.4	3.4	-6.2	-11.4
	DBStd	12.30	10.21	8.44	8.21	5.19	4.57	3.12	2.99	3.44	4.35	5.10	7.57	9.10
	HDD10.0	3408	703	551	480	204	57	4	1	4	47	209	485	662
	HDD18.3	5872	962	784	738	451	261	131	77	112	239	463	735	921
	CDD10.0	603	0	0	0	3	54	127	196	158	59	5	0	0
	CDD18.3	26	0	0	0	0	1	3	13	8	1	0	0	0
	CDH23.3	293	0	0	0	1	14	43	128	93	14	1	0	0
	CDH26.7	42	0	0	0	0	1	4	22	14	1	0	0	0
Wind	WSAvg	3.9	4.0	3.8	4.0	4.2	4.0	3.7	3.6	3.5	3.8	4.3	4.0	3.9
Precipitation	PrecAvg	477	30	22	25	21	47	70	75	60	45	29	29	28
	PrecMax	601	77	64	53	51	112	147	167	136	129	66	58	95
	PrecMin	330	6	4	1	5	4	14	14	7	4	4	3	4
	PrecStd	76	19	15	15	14	27	33	44	31	31	15	14	18
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.9	7.2	11.1	18.9	25.2	26.9	29.1	28.4	25.2	18.4	9.6	7.2
		MCWB	2.4	2.8	5.2	9.3	13.4	16.1	17.5	16.8	14.7	10.4	5.2	3.4
	2%	DB	5.0	4.9	8.0	15.5	22.4	24.4	26.4	25.8	22.3	14.5	6.7	5.1
		MCWB	1.9	1.3	3.4	7.4	11.8	14.5	16.5	16.4	13.7	8.2	3.3	1.9
	5%	DB	3.5	3.3	6.2	12.9	20.0	22.6	24.5	23.8	19.7	12.2	4.8	3.3
		MCWB	0.8	0.1	2.0	5.8	10.7	13.7	15.8	15.5	12.3	7.0	1.8	0.8
	10%	DB	1.7	1.6	4.6	10.9	17.9	20.7	22.7	21.9	17.5	10.3	3.2	1.1
		MCWB	-0.6	-1.1	1.0	4.6	9.7	12.7	15.0	14.6	11.1	5.7	0.6	-1.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.4	3.3	5.5	9.9	14.8	17.5	19.2	19.0	15.9	11.4	5.5	3.7
		MCDB	6.4	6.5	10.0	17.9	22.3	24.7	25.7	24.6	23.1	17.4	9.0	6.6
	2%	WB	2.1	1.6	3.6	7.8	12.8	15.7	17.6	17.4	14.1	9.0	3.6	2.1
		MCDB	4.6	4.5	7.6	14.6	20.3	22.5	24.3	23.8	21.0	13.9	6.4	4.8
	5%	WB	0.9	0.4	2.3	6.2	11.4	14.5	16.5	16.2	12.7	7.3	2.0	0.8
		MCDB	3.4	3.1	5.8	12.0	18.5	20.9	23.0	22.5	18.7	11.6	4.2	3.2
	10%	WB	-0.7	-1.1	1.1	5.0	10.2	13.4	15.5	15.1	11.6	6.0	0.8	-1.0
		MCDB	1.7	1.6	4.4	10.3	16.9	19.4	21.5	20.7	16.7	10.0	3.0	1.0
Mean Daily Temperature Range	5% DB	MDBR	7.2	7.7	8.3	9.4	10.7	10.1	10.2	10.5	9.8	7.4	6.3	6.9
		MCDDBR	7.2	8.5	9.0	12.8	13.8	13.0	13.2	13.6	13.2	10.4	7.2	7.4
		MCWBR	5.6	6.3	5.9	7.1	6.9	6.1	6.0	6.2	6.8	6.1	5.4	5.9
	5% WB	MCDWBR	7.6	8.4	9.1	12.0	12.9	12.0	12.5	12.6	12.6	10.1	7.2	7.7
		MCWBR	6.1	6.4	6.3	7.0	6.7	6.0	6.1	6.3	6.8	6.4	5.6	6.4
Clear-Sky Solar Irradiance	taub	0.233	0.248	0.274	0.318	0.344	0.344	0.344	0.341	0.307	0.282	0.239	0.218	
	taud	2.301	2.351	2.330	2.376	2.404	2.408	2.446	2.451	2.548	2.569	2.404	2.242	
	Ebn at Noon	711	847	902	904	896	898	891	871	862	804	706	628	
	Edh at Noon	48	69	91	103	107	108	102	95	75	57	43	39	
All-Sky Solar Radiation	RadAvg	0.64	1.55	2.98	4.67	5.49	6.03	5.84	4.74	3.22	1.69	0.76	0.46	
	RadStd	0.07	0.12	0.20	0.21	0.38	0.54	0.33	0.42	0.29	0.18	0.07	0.03	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (13 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page